

Analysis of Social Barriers, Digital Access, and Internal Bottlenecks

Immortal | Hillal Hariyono, Hans Richard Djojasantoso, Ryan Christopher Setiawan, Vinson Nicholas Sorensen

~5x Higher ED Rate with Barriers 40.1% vs 8.4% no-barrier journeys	34.6 visits Avg Visits — Patient Declined vs 23.8 Activated (50% more)	1,252 pts Patients per Provider Emergency Medicine — ~5x P75 threshold
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Background & Problem Statement

Every year, millions of Americans end up in the Emergency Room simply because they had no other choice. This project uses four years of real patient data from **Stormont Vail Health (SVH)** to answer a core question: do patients end up in the ED because their condition suddenly worsens — or because **social barriers leave them with no other option?** Analysis across encounter, patient, and social determinant data surfaces two dominant barriers — **Transportation Needs** and **Financial Resource Strain** — that redirect patients away from affordable clinic care and toward the ED as their only viable access point.

Methodology

① Social Barriers & Patient Journeys Patients classified into No Barrier, Financial Only, Transport Only, and Both Barriers. Barrier–ED relationship validated against ACS Census vehicle-ownership data (block group & tract level).	② The Digital Gap MyChart status profiled by average visit count, birth-year cohort, and top diagnosis mix to test whether digital disengagement correlates with highest clinical burden.	③ System Bottleneck Mapping Patient-to-provider ratios and inter-visit gap days computed across high-volume specialties (≥1,000 encounters). P75 threshold used to flag bottleneck specialties. ED outflow mapped as a directed graph, gap-day distributions inspected for the Emergency Medicine Specialty.
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Evaluation

Social Barriers & Patient Journeys: Barrier patients are not dropping out — they are being redirected. Journeys for transport-barrier patients average **3.46 encounters** versus **2.31 for no-barrier patients** — 60% more encounters for the most vulnerable group. Though only 0.8% of all journeys carry both barriers, **40.1% of barrier journeys included an ED visit** versus just 8.4% for no-barrier journeys — nearly 5 times higher. ACS Census external validation confirms the barrier is real: transport self-reports rise from 2.9% in well-connected neighborhoods to **9.8%** where >15% of households lack a car (3.3× higher). The top diagnoses — general adult medical examination (#1), essential hypertension (#2), long-term insulin use (#3), Type 2 diabetes, and major depressive disorder — are all chronic conditions requiring consistent management, not emergencies. Home mapping pinpoints **Harvey, Russell, and Butler Counties** as the highest-concentration areas.

The Digital Gap: Patients who explicitly declined MyChart average **34.6 visits over 3 years** — nearly 50% more than activated users (23.8). They are not non-users; they are power users who show up in person every time. The age profile is dominated by patients born between **1930 and 1965**, presenting repeatedly for the same chronic conditions seen in the barrier analysis (essential hypertension: 2,777 cases; general exam: 1,688; insulin use: 1,308). This cohort is entirely unreachable through digital channels.

System Bottlenecks: Emergency Medicine sits at **1,252 patients per provider** and Pediatric Hospitalist at 1,228 — both nearly 5× the P75 threshold of 267. The overall Pearson correlation of load vs. wait time is a weak 0.06, but only because most departments are manageable; the top-5 loaded specialties cluster tightly in the high-load, high-delay danger zone. After an ED visit, almost every outflow edge exceeds 14 days: OR averages 204 days, Labor & Delivery 248 days, Normal Outpatient 115 days. The gap-day boxplot shows Emergency Medicine's middle 50% spanning 0–85 days. Vulnerable patients managing diabetes or hypertension face three-month gaps that are clinically dangerous, not merely inconvenient.

Conclusion & Application

The data tells one coherent story: vulnerable patients are trapped (redirected to the ED by social barriers), invisible to digital outreach, then stuck behind specialty bottlenecks with no fast exit. Three targeted interventions address each failure point. **(1) Proactive chronic disease outreach** concentrated in Harvey, Russell, and Butler Counties — reaching patients before they deteriorate and arrive by ambulance. **(2) Non-digital communication pathways** — phone-based reminders and physical mail for elderly patients born 1930–1965 who declined MyChart, the group that visits more than

anyone else. **(3) ED bottleneck relief** — expand Emergency Medicine capacity and build fast-track referral pathways so that post-ED wait times of 204–248 days become the exception, not the rule.